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Tiled apparatuses

Abstract

A tiled apparatus is provided by the present disclosure. at least two of the tiled apparatus include display panels, backlight modules and lens parts. the display panel includes a display part and a tiled part. The tiled part is composed of transparent materials. An orthographic projection of the backlight module on the display panel covers the display part and at least a part of the tiled part. The lens part is disposed on a side of the display panel away from the backlight module and covers adjacent two tiled parts and a seam. The lens part partially overlaps with display parts of adjacent two display panels.

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Background/Summary**CROSS REFERENCE TO RELATED APPLICATIONS**

(1) This application is a US national phase application based upon an International Application No. PCT/CN2023/094504, filed on May 16, 2023, which claims priority to Chinese Patent Application No. 202310461668.5, filed on Apr. 19, 2023. The entire disclosures of the aforementioned applications are incorporated herein by reference in their entireties.

TECHNICAL FIELD

(2) The present disclosure relates to a field of display technologies, and in particular to tiled

apparatuses.

BACKGROUND

(3) At present, in order to achieve large-scale display of display devices, display splicing technology is rapidly developing. However, the tiled devices for organic light-emitting diode display panels and micro diode direct display panels cannot be widely promoted due to problems such as cost and reliability. Therefore, tiled devices for liquid crystal display panels are still mainstream products on the market.

(4) In existing liquid crystal display tiled devices, seams between display screens are usually eliminated by electrical methods. However, an image brightness in a tiled area of the tiled display device by the technical solution is lower than that in a non-tiled area, leading to a technical problem of uneven display in the tiled device.

SUMMARY

(5) The present disclosure provides a tiled apparatus to improve the technical problem of uneven display caused by inconsistent display brightness between a tiled area and a non-tiled area in the existing tiled device.

(6) The present disclosure provides a tiled apparatus including at least two display devices tiled with each other. A seam is defined between adjacent two display devices, and each display device includes: a display panel including a display part and a tiled part located on a side of the display part, wherein tiled parts of the adjacent two display devices are adjacent to each other, and the tiled parts are composed of transparent materials; and a backlight module disposed on a side of the display panel away from a display surface, wherein an orthographic projection of the backlight module on the display panel covers the display part and at least a part of the tiled part, and light emitted from an area of the backlight module corresponding to the tiled part is capable of passing through the tiled part.

(7) The tiled apparatus further includes a lens part disposed on a side of the display panel away from the backlight module, the lens part covers adjacent two tiled part and the seam, and the lens part overlaps with a part of display parts of adjacent two display panels.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a top structural view of the tiled apparatus provided by the present disclosure.

(2) FIG. 2 is a first cross-sectional view of a section AA in FIG. 1.

(3) FIG. 3 is a cross-sectional view of a display panel of the tiled apparatus provided by the present disclosure.

(4) FIG. 4 is a second cross-sectional view of the section AA in FIG. 1.

(5) FIG. 5 is a third cross-sectional view of the section AA in FIG. 1.

(6) FIG. 6 is an optical path structure view of a Fresnel lens of the tiled apparatus provided by the present disclosure.

(7) FIG. 7 is a fourth cross-sectional view of the section AA in FIG. 1.

(8) FIG. 8 is a fifth cross-sectional view of the section AA in FIG. 1.

(9) FIG. 9 is a sixth cross-sectional view of the section AA in FIG. 1.

(10) FIG. 10 is a seventh cross-sectional view of the section AA in FIG. 1.

DETAILED DESCRIPTION

(11) To make the objectives, technical solutions, and effects of the present disclosure clearer and more specific, the present disclosure is described in further detail below with reference to the embodiments accompanying with drawings. It should be understood that the specific embodiments described herein are merely for explaining the present disclosure, and the present disclosure is not limited thereto.

(12) In existing tiled devices, seams between display screens are usually eliminated by optical or electrical methods. However, the technical solution causes an image brightness in a tiled area to be lower than that in a non-tiled area, resulting in uneven display in the tiled apparatus. The present disclosure provides the following technical solutions to solve the foregoing technical problems.

(13) Referring to FIGS. 1 to 10, the present disclosure provides a tiled apparatus **100**, which includes at least two display devices **200** tiled with each other. A seam **130** is defined between adjacent two display devices **200**. Each of the display devices **200** includes a display panel **10** and a backlight module **20** disposed on a side of the display panel **10** away from a display surface.

(14) In the embodiment, the display panel **10** includes a display part **110** and a tiled part **120** located on a side of the display part **110**. The tiled parts **120** of the two adjacent display devices **200** are adjacent to each other, and the tiled parts **120** are composed of transparent materials.

(15) In the embodiment, an orthographic projection of the backlight module **20** on the display panel **10** covers the display part **110** and at least a part of the tiled part **120**, and light emitted from an area of the backlight module **20** corresponding to the tiled part **120** is capable of passing through the tiled part **120**. The backlight module **20** includes a plurality of light-emitting component, and the light emitted by the light-emitting component is capable of passing through the transparent materials of the tiled part **120** and being derived from a light emitting surface of the display panel **10**.

(16) In the embodiment, the tiled apparatus **100** further includes a lens part **30**, which is disposed on a side of the display panel **10** away from the backlight module **20**. The lens part **30** covers adjacent two tiled parts **120** and the seam, and the lens part **30** partially overlaps with the display part **110** of adjacent two display panels **10**.

(17) The tiled part **120** of the display panel **10** in the present disclosure is disposed to be transparent, so that the light emitted by the backlight module **20** corresponding to the tiled part **120** is capable of passing through the tiled part **120**. And the lens part **30** guides the light passing through the tiled part **120** and part of display parts **110** to the tiled parts **120** and the seam **130** between adjacent tiled parts **120**. While ensuring the continuity of the displayed image, the image brightness in a tiled area PP is improved, and the technical problem of uneven display of the tiled apparatus **100** is solved.

(18) In the embodiment, the tiled part **120** is only equipped with transparent materials, and relevant metal wirings or non-transparent materials in the display panel may be disposed at a boundary of the non-tiled area.

(19) In the embodiment, the backlight module **20** may include a plurality of light-emitting component, and the light emitted by the plurality of the light-emitting component is incident to the tiled parts **120** and the display parts **110** by optical diaphragms in the backlight module **20**.

(20) In the embodiment, the backlight module **20** may be a straight down backlight module or a side entry backlight module according to the position of the light-emitting component. When the backlight module **20** is the straight down backlight module, the light emitted from an area of the backlight module **20** corresponding to the tiled part **120** is capable of passing through the tiled parts **120** and enter the lens part **30**. When the backlight module **20** is the side entry backlight module, a light guide plate of the backlight module **20** extends below the tiled part **120**, the light guide plate and the tiled part **120** are overlapped, and the light emitted by the light-emitting component is guided to the tiled area PP through the light guide plate and incident to the tiled parts **120**.

(21) In the embodiment, the light-emitting component may be light-emitting diodes (LEDs), Mini LEDs, Micro LEDs, etc.

(22) The technical solutions of the present disclosure are described in conjunction with specific embodiments.

(23) The tiled apparatus **100** in the embodiment may be provided to be two display devices **200** tiled opposite to each other, four display devices **200** tiled in a Chinese character shape of “custom character”, or by other numbers of display devices **200** for tiling. The embodiment

illustrates the technical solution of the present disclosure by two display devices **200** tiled relatively with each other.

(24) Referring to FIGS. **1** and **2**. The tiled apparatus **100** may include two display devices **200** tiled opposite to each other, and structures of the two display devices **200** may be the same. The tiled apparatus **100** may include the tiled area PP and the non-tiled area NP. The tiled area PP is provided with tiled parts **120** of the two display panels **10** and a seam **130** located between the two tiled parts **120**, and the non-tiled area NP is provided with display parts **110** of the two display panels **10**.

(25) Referring to FIG. **3**. The display panel **10** may include a first substrate **510**, a second substrate **520** opposite to the first substrate **510**, and a liquid crystal layer **530** between the first substrate **510** and the second substrate **520**. The first substrate **510** may be a conventional array substrate or a color filter on array (COA) substrate. In the following embodiment, the present disclosure takes the COA substrate as an example for explanation.

(26) In the embodiment, the first substrate **510** may include a first substrate **511**, a thin film transistor layer located on the first substrate **511**. A material of the first substrate **511** may be made of glass, quartz or polyimide.

(27) In the embodiment, the thin film transistor layer may include a plurality of thin film transistors. The thin film transistor may be an etching barrier type, a back channel etching type. or divided into structures such as a bottom gate thin film transistor, a top gate thin film transistor, etc. based on the position of the gate and active layer. For example, the thin film transistor which is the bottom gate thin film transistor type may include a gate layer **512** located on the first substrate **511**, a gate insulation layer **513** located on the gate insulation layer **512**, an active layer **514** located on the gate insulation layer **513**, a source drain layer **515** located on the active layer **514**, a color resistance layer **516** located on the source drain layer **515**, and a pixel electrode layer **517** located on the color resistance layer **516**. In other embodiments, a flat layer or other insulation film layers are furtherly disposed between the source drain layer **515** and the color resistance layer **516**, which is not limited here. Secondly, other transparent insulation film layers such as the gate insulation layer **513** may extend from the display part **110** to the tiled part **120**.

(28) In the embodiment, the second substrate **520** may include a second substrate **521** and a common electrode layer **522** located on the second substrate **521**. The electric field formed by the common electrode layer **522** and the pixel electrode layer **517** drives a deflection of liquid crystal molecules in the liquid crystal layer **530**.

(29) In the embodiment, a plurality of second main spacer columns **518** and sub spacer columns (not shown in the FIG.) are further provided between the first substrate **510** and the second substrate **520**. A thickness of the sub spacer column is less than that of the second main spacer column **518**, and the second main spacer column **518** is connected to the first substrate **510** and the second substrate **520**.

(30) In the tiled apparatus **100** of the present disclosure, referring to FIGS. **1** and **2**, the display device **200** further includes a first polarizer **540** and a second polarizer **550** disposed on both sides of the display panel **10**. The first polarizer **540** may be disposed between the display panel **10** and the backlight module **20**, and the second polarizer **550** may be disposed between the display panel **10** and the lens part **30**. At most one of the first polarizer **540** and the second polarizer **550** overlaps with the tiled part **120**.

(31) As the display panel **10** of the present disclosure is a liquid crystal panel, in order to achieve image display of the liquid crystal display panel, an upper polarizer and a lower polarizer are disposed on both sides of the liquid crystal panel. However, since the tiled part **120** is composed of transparent materials, and no corresponding thin film transistor is disposed on the tiled part **120** to control the deflection of the liquid crystal in the tiled part **120**, when the upper polarizer and the lower polarizer are disposed on both sides of the tiled part **120**, the polarized light entering the display panel **10** is not regulated by the liquid crystal, and the polarized light cannot pass through the upper polarizer, resulting in the inability of the light incident on the tiled part **120** to enter the

lens part **30**, which cannot solve the technical problem of inconsistent brightness between the tiled area PP and the non-tiled area NP.

(32) In the embodiment, at most one of the first polarizer **540** and the second polarizer **550** of the present disclosure is provided to overlap with the tiled part **120**. For example, referring to FIG. **4**, the first polarizer **540** and the tiled part **120** are overlapped, and the light incident on the tiled part **120** passes through the first polarizer **540** to form polarized light. Since the second polarizer **550** and the tiled part **120** are not overlapped, even if the liquid crystal does not regulate the polarized light entering the display panel **10**, the polarized light is capable of entering the lens part **30**, to improve the technical problem of inconsistent brightness between the tiled area PP and the non-tiled area NP. Similarly, referring to FIG. **1**, the second polarizer **550** and the tiled part **120** are overlapped, and the first polarizer **540** and the tiled part **120** are not overlapped. Alternatively, referring to FIG. **5**, neither the first polarizer **540** nor the second polarizer **550** overlaps with the tiled part **120**, allowing light incident to the tiled part **120** to enter the lens part **30** to improve the technical problem of inconsistent brightness between the tiled area PP and the non-tiled area NP.

(33) In the embodiment, referring to the structure show in FIG. **2**, the second polarizer **550** and the tiled part **120** are overlapped, and the first polarizer **540** and the tiled part **120** are not overlapped. The disposition of the second polarizer **550** may remove the ambient light shining on the tiled area PP and reduce the interference of ambient light on the display brightness of the tiled area PP.

(34) In the tiled apparatus **100** of the present disclosure, referring to FIGS. **2** to **6**. The lens part **30** may include at least one lens in a strip shape and with a convex lens structure. The lens may converge the light emitted from the display part **110** and the tiled part **120** to the lens to a focal area of the lens, therefore the image brightness in the tiled area PP is improved while ensuring the continuity of the displayed image, and the technical problem of uneven display in the tiled apparatus **100** are improved.

(35) It should be noted that the lens may be a conventional convex lens or a Fresnel lens **310**. Taking the Fresnel lens **310** as an example to explain the technical solutions of the present disclosure in the following description.

(36) Referring to FIGS. **2** to **6**. The lens may be the Fresnel lens **310**, which has a convex lens structure **311** and at least one toothed structure **312** disposed on both sides of the convex lens structure **311**. At least one toothed structure **312** partially overlaps with the display part **110** of a corresponding display panel **10**.

(37) In the structure of FIG. **6**, the Fresnel lens **310**, also known as a threaded lens, is made of a thin sheet of polyolefin material injected or made of glass. One side of the Fresnel lens **310** is a plane, and the other side is removed as much optical material as possible to reduce a thickness and retain surface curvature, forming a plurality of the toothed structures **312** on both sides of the convex lens structure **311**. The conventional Fresnel lens has several annular toothed structures disposed in a concentric circle, and a convex lens structure surrounded by the annular toothed structures. In the present disclosure, in order to guide the image of the non-tiled area NP to the tiled area PP, the Fresnel lens **310** is arranged in a long strip shape, that is, each cross-sectional of the Fresnel lens **310** is the structure shown in FIG. **6**.

(38) In the structure of FIG. **6**, after the light incident on the tiled part **120** enters the Fresnel lens **310**, due to the action of the plurality of the toothed structures **312** and the convex lens structure **311**, the incident light may converge to the focal point of the Fresnel lens **310**.

(39) In the tiled apparatus **100** of the present disclosure, referring to FIGS. **2** and **4**. The lens part **30** may include a Fresnel lens **310** in a strip shape, which simultaneously covers the tiled parts **120** of the two display panels **10** and the seam **130**. A part of the first toothed structures **312** on a first side of the Fresnel lens **310** overlaps with the tiled part **120** on one side, and a part of the second toothed structures **312** on a second side of the Fresnel lens **310** overlaps with the tiled part **120** on the other side. The convex lens structure **311** of the Fresnel lens **310** covers parts of the tiled parts **120** of the two display panels **10** and the seam **130**. The overlapping areas of the toothed structure **312** on both

sides of the Fresnel lens **310** and the corresponding display part **110** are the same.

(40) In an embodiment, an orthographic projection of a focus of the Fresnel lens **310** on the tiled apparatus **100** is located in the seam **130**. The light passing through the tiled parts **120** and parts of the display parts **110** converges towards the focal area of the Fresnel lens **310** by the Fresnel lens **310**, improving the image brightness in the tiled area PP while ensuring the continuity of the displayed image, and improving the technical problem of uneven display of the tiled apparatus **100**.

(41) In the tiled apparatus **100** of the present disclosure, the lens part **30** includes at least two Fresnel lenses **310** connected to each other, at least one Fresnel lens **310** corresponds to a display panel **10**. An orthographic projection of the focus of the Fresnel lens **310** on the tiled apparatus **100** is located in the tiled part **120** of a corresponding display panel **10**. In the structure of FIG. 5, the lens part **30** includes two Fresnel lenses **310** connected to each other as an example for explanation.

(42) In an embodiment, the two Fresnel lenses **310** are disposed in a strip shape along boundaries of corresponding display panels **10**. The two Fresnel lenses **310** are symmetrically disposed with an axis plane O of the two display panels **10** as a symmetry plane, and connection points of the adjacent two Fresnel lenses **310** are located in the axis plane O.

(43) In the embodiment, the light passing through the tiled part **120** and a part of the display part **110** converges towards the focal area of the Fresnel lens **310** by the corresponding Fresnel lens **310**. While ensuring the continuity of the displayed images in the two display panels **10**, the image brightness in the tiled area PP is improved, and the technical problem of uneven display in the tiled apparatus **100** is improved.

(44) In the tiled apparatus **100** of the present disclosure, referring to FIGS. 2, 4, and 5, the display part **110** includes a plurality of sub pixels, and the lens part **30** covers at least three sub pixels arranged along a first direction, which is parallel to a direction from the display part **110** pointing to the tiled part **120**.

(45) In an embodiment, the tiled part **120** is not provided with a corresponding color resistance unit. In order to display image in the tiled area PP, it needs to enlarge the image of the boundary of the non-tiled area NP to the tiled area PP, so that visual continuity of the image between the adjacent two display devices **200** is capable of being achieved. The lens part **30** of the present disclosure overlaps with at least three sub pixels of the display part **110**, and the three sub pixels are red sub pixels, green sub pixels, and blue sub pixels. The image of the at least three sub pixels at the boundary of the display part **110** is directed to the tiled area PP to achieve continuous image of the tiled area PP of the tiled apparatus **100**.

(46) In the embodiment, along the first direction, an overlapping dimension L between the lens part **30** and the display part **110** is less than or equal to 1 mm. If the lens part **30** covers too many sub pixels, it will affect the image quality of the display part **110**. Therefore, the overlap area between the lens part **30** and the display part **110** cannot be too large. A maximum overlap size L of the present disclosure is limited to be 1 mm, so that a large-sized tiled apparatus **100** is capable of achieving visual continuity in the image of the tiled area PP.

(47) In the tiled apparatus **100** of the present disclosure, referring to FIGS. 2, 4, and 5. The tiled apparatus **100** further includes a protective layer **40** located on a side of the display panel **10** away from the backlight module **20**. The protective layer **40** covers the lens part **30**, and a thickness of the protective layer **40** is greater than or equal to a maximum thickness of the lens part **30**.

(48) In an embodiment, the protective layer **40** may cover the display parts **110** and the tiled parts **120**. Surfaces of the protective layer **40** away from the backlight module **20** are on the same plane, and distances between the surfaces of the protective layer at different positions on one side away from the backlight module and the backlight module are equal.

(49) In the embodiment, the protective layer **40** is mainly used to protect the lens parts **30** and prevent surfaces of the lens parts **30** from being worn by upper hard film layers. Additionally, the protective layer **40** may further ensure the flatness of the surface of the tiled apparatus **100**.

(50) In the embodiment, a material of the protective layer **40** may be an organic material such as

polymethyl methacrylate.

(51) Based on of the above embodiments, referring to FIG. 7, the tiled apparatus **100** may further include a transparent optical adhesive **131** disposed between the two adjacent display panels **10**, and the transparent optical adhesive **131** fills the seam **130** between the two display panels **10**.

(52) In an embodiment, due to the certain loss rate of the light passing through transparent materials, and the fact that no material is disposed in the seam **130** when the light passes through the seam **130**, the amount of light entering the lens **30** is different when the light passes through the seam **120** and the seam **130**, resulting in inconsistent display brightness between the seam **130** and the seam **120**. The present disclosure balances the transmittance of light passing through the tiled part **120** and the seam **130** by disposing the transparent optical adhesive **131** in the seam **130**, and improves the technical problem of inconsistent display brightness between the seam **130** and the tiled part **120**.

(53) In the embodiment, a refractive index of the transparent optical adhesive **131** may be the same as that of the first substrate **511** or/and the second substrate **521**. Angles at which light enters or exits the substrate are the same, resulting in a smooth transition of the display pictures of the two display panels **10**. In addition, the seam **130** is filled with the transparent optical adhesive **131**, which is capable of preventing dust or other pollutants during the installation process from entering the tiled display device **200**.

(54) In the embodiment, a frame adhesive **150** may be composed of a transparent adhesive material, and an absolute value of difference between a light transmittance of the frame adhesive **150** and a light transmittance of the transparent optical adhesive **131** may range from 0 to 5%.

(55) In the tiled apparatus **100** of the present disclosure, referring to the structures of FIGS. 2 and 3. The first polarizer **540** and the second polarizer **550** are generally disposed in an orthogonal manner. At a side view angle of the tiled apparatus **100**, the light emitted by the light-emitting component in the display part **110** enters the tiled part **120** through the first polarizer **540**, and passes through the second polarizer **550** in the tiled area PP, so that the user may observe the bright line in the tiled area PP, which causing a technical problem of light leakage at the side view angle of the tiled apparatus **100**. The existing technology usually adds a compensation film in the first polarizer **540** and the second polarizer **550** to improve the problem of light leakage at the side view angle. However, the compensation film usually needs to be matched with a liquid crystal gap of the display part **110**. Therefore, a liquid crystal gap in the tiled area PP needs to match a liquid crystal gap in the tiled part **120** to improve the problem of light leakage at the side view angle.

(56) In the tiled apparatus **100** of the present disclosure, referring to FIG. 3, the tiled part **120** includes the frame adhesive **150** and a support part **140**. The frame adhesive **150** is disposed at a boundary of the display panel **10**, and the support part **140** is disposed between the frame adhesive **150** and the display part **110**. The frame adhesive **150** and the support part **140** are disposed between the first substrate **511** and the second substrate **521**.

(57) In an embodiment, the support part **140** includes support body **141** in a strip shape and a plurality of first main spacer columns **142** disposed on the support body **141**. In a top view direction of the tiled apparatus **100**, an area of the first main spacer column **142** is smaller than that of the support body **141**.

(58) In the embodiment, a material of the first main spacer column **142** and a material of the support body **141** may be the same and may be composed of transparent flexible materials.

(59) In the embodiment, without disposing the support part **140**, an area between the display part **110** and the frame adhesive **150** is filled with liquid crystals, and the liquid crystal gap in the area is larger than the liquid crystal gap in the display part **110**. Therefore, the liquid crystal gap in the area does not match the liquid crystal gap in the display part **110**. Referring to FIG. 3. The support body **141** in the present disclosure is in a strip shape, and the liquid crystal gap M2 in an area where the support body **141** is located is smaller than the liquid crystal gap in the area where the support body **141** is not provided, so that the liquid crystal gap M2 in the area where the support body **141** is

located is close to or equal to the liquid crystal gap M1 in the display part **110**, solving the technical problem of light leakage at the side view angle of the tiled apparatus **100**.

(60) In the embodiment, the plurality of the first main spacer columns **142** are connected to the support bodies **141** and the second substrate **521**, and the plurality of the first main spacer columns **142** are mainly used to adjust a spacing of a liquid crystal cell in the area to avoid the spacing of the liquid crystal cell in the area being smaller than a spacing of a liquid crystal cell in the display part **110**.

(61) In the tiled apparatus **100** of the present disclosure, referring to FIG. **3**, the display part **110** further includes a second main spacer column **518**, and an absolute value of difference between a thickness of the first main spacer column **142** and a thickness of the second main spacer column **518** ranges from 0 to 0.2 microns.

(62) In the embodiment, the thickness of the first main spacer column **142** and the thickness of the second main spacer column **518** may be equal.

(63) In the embodiment, the first main spacer column **142** and the second main spacer column **518** may be formed by a same mask process, simplifying the manufacturing process while ensuring the liquid crystal cell spacing of the tiled part **120**. In addition, the thickness of the first main spacer column **142** is equal to the liquid crystal gap in the area where the support body **141** is located, and the thickness of the second main spacer column **518** is equal to the liquid crystal gap in the display part **110**. Therefore, the thickness of the first main spacer column **142** of the present disclosure is controlled to make the liquid crystal gap in the area where the support body **141** is located equal to the liquid crystal gap in the display part **110**. The technical problem of light leakage at the side view angle of the tiled apparatus **100** is solved.

(64) In the embodiment, the second main spacer column **518** may be disposed on structures in a non-transparent area such as a thin film transistor or gate line to avoid the second main spacer column **518** affecting the transmittance of a transparent area of the display panel **110**. The second main spacer column **518** is mainly used to adjust the liquid crystal cell spacing of the display panel **110**.

(65) In the embodiment, since the relevant metal wirings or non-transparent materials of the present disclosure may be disposed at the boundary of the non-tiled area, and the tiled part is only composed of transparent materials, a size of the tiled part of the present disclosure may be less than 300 microns. In addition, due to the fact that the image of the tiled part is directed towards the tiled part after being enlarged by the image of the display part, if the size of the tiled part is too large, the image resolution of the area is too low, and it cannot achieve visual continuity with the high-resolution image of the display part. Therefore, the size of the tiled part in the present disclosure may be less than or equal to 3 mm.

(66) Since the light entering the Fresnel lens **310** converges to the focus area of the Fresnel lens **310**, that is, the light entering the seam **130** comes from the light passing through the tiled part **120** and part of the display part **110**, and the position corresponding to the seam **130** in the backlight module is not provided with a light-emitting component **220**, so the image brightness of the seam **130** is less than the brightness of the area around the seam **130**, causing uneven display in the tiled area PP.

(67) Referring to FIG. **8**, The backlight module **20** may be a straight down backlight module, and the backlight module **20** may include a light-emitting component **220** and a light compensating component **210**. In the top view direction of the tiled apparatus **100**, the light-emitting component **220** overlaps with the display part **110**, and the light compensating component **210** overlaps with the tiled part **120**.

(68) In an embodiment, a plurality of the light compensating components **210** are disposed on the backlight module **20** in the present disclosure, which ensures that the light emitted by the light compensating components **210** may directly enter the tiled part **120**, and the light incident on the tiled part **120** may be converged to the area where the seam **130** is located with the help of the lens

part **30**. The technical problem of inconsistent brightness between the seam **130** and the seam **120** in the tiled area PP is improved.

(69) In the embodiment, the light compensating component **210** may be Micro LEDs, and the light-emitting component **220** may be LEDs, Mini LEDs or Micro LEDs.

(70) Referring to FIG. **9**, the backlight module **20** may be a straight down backlight module, and the backlight module **20** may include a light-emitting component **220** and a diffusion plate **230**, and the diffusion plate **230** is disposed on a side of the light-emitting component **220** close to the display panel **10**. In the top view direction of the tiled apparatus **100**, the light-emitting component **220** overlaps with the display part **110**, and the diffusion plate **230** extends to an area where the tiled part **120** is located.

(71) In an embodiment, the diffusion plate **230** extends to the area where the tiled part **120** is located in the present disclosure, so that part of the light emitted by the light-emitting component **220** overlapping with the display part **110** is capable of diffusing to the tiled part **120** through the diffusion plate **230**. Combining with the lens part **30**, the light incident on the tiled part **120** is converged to the area where the seam **130** is located, and the technical problem of inconsistent brightness between the seam **130** and tiled part **120** in the tiled area PP is solved.

(72) In the embodiment, the light-emitting component **220** may be LEDs, Mini LEDs or Micro LEDs.

(73) Referring to FIG. **10**, the backlight module **20** may be a side entry backlight module, and the backlight module **20** may include a light-emitting component **220** and a light guide plate **240**. In the top view direction of the tiled apparatus **100**, the light guide plate **240** extends from an area where the display part **110** is located to an area where the tiled part **120** is located.

(74) In an embodiment, the light guide plate **240** extends from the area where the display part **110** is located to the area where the tiled part **120** is located in the present disclosure, so that the light emitted by the light-emitting component **220** is capable of being imported to the tiled part **120** through the light guide plate **240**. Combining with the lens part **30**, the light incident on the tiled part **120** is converged to the area where the seam **130** is located, and the technical problem of inconsistent brightness between the seam **130** and tiled part **120** in the tiled area PP is solved.

(75) In the embodiment, the light-emitting component **220** may be LEDs or Mini LEDs.

(76) In the above embodiment, the size of the LED is in millimeters. The size of the Mini LED is in micrometers, such as ranging from 50 microns to 200 microns. The size of the Micro LED is in nanometers, such as less than 10 microns.

(77) It can be understood that, for those skilled in the art, equivalent replacements and modifications can be made according to the technical solution and disclosure ideas thereof of the present disclosure, and all these modifications or replacements are considered within the protection scope of the attached claims of the present disclosure.

Claims

1. A tiled apparatus, comprising at least two display devices tiled with each other, wherein a seam is defined between adjacent two display devices, and each display device comprises: a display panel comprising a display part and a tiled part located on a side of the display part, wherein tiled parts of the adjacent two display devices are adjacent to each other, and the tiled parts are composed of transparent materials; and a backlight module disposed on a side of the display panel away from a display surface, wherein an orthographic projection of the backlight module on the display panel covers the display part and at least a part of the tiled part, and light emitted from an area of the backlight module corresponding to the tiled part is capable of passing through the tiled part; wherein the tiled apparatus further comprises a lens part disposed on a side of the display panel away from the backlight module, the lens part covers adjacent two tiled parts and the seam, and the lens part partially overlaps with display parts of adjacent two display panels; wherein the

lens part comprises at least one lens in a strip shape and with a convex lens structure.

2. The tiled apparatus of claim 1, wherein the display device further comprises a first polarizer and a second polarizer disposed on both sides of the display panel, and at most one of the first polarizer and the second polarizer overlaps with the tiled part.

3. The tiled apparatus of claim 1, wherein the lens is a Fresnel lens, the Fresnel lens comprises the convex lens structure and at least one toothed structure disposed on both sides of the convex lens structure, and the at least one toothed structure partially overlaps with the display part of a corresponding display panel.

4. The tiled apparatus of claim 1, wherein the lens part comprises one lens, and an orthographic projection of a focus of the lens on the tiled apparatus is located in the seam.

5. The tiled apparatus of claim 1, wherein the lens part comprises at least two lenses connected to each other, at least one lens corresponds to a display panel, and an orthographic projection of a focus of the lens on the tiled apparatus is located in the tiled part of a corresponding display panel.

6. The tiled apparatus of claim 1, wherein the display part comprises a plurality of sub pixels, and the lens part covers at least three of the sub pixels disposed along a first direction; and wherein the first direction is parallel to a direction from the display part pointing to the tiled part.

7. The tiled apparatus of claim 1, wherein the tiled apparatus further comprises a protective layer disposed on a side of the display panel away from the backlight module, the protective layer covers the display part and the tiled part, a thickness of the protective layer is greater than or equal to a maximum thickness of the lens part, surfaces of the protective layer away from the backlight module are on a same plane, and distances between the surfaces of the protective layer at different positions on one side away from the backlight module and the backlight module are equal.

8. The tiled apparatus of claim 7, wherein a material of the protective layer is polymethyl methacrylate.

9. The tiled apparatus of claim 1, wherein the backlight module is a straight down backlight module, which comprises a light-emitting component and a light compensating component; and wherein in a top view direction of the tiled apparatus, the light-emitting component overlaps with the display part, and the light compensating component overlaps with the tiled part.

10. The tiled apparatus of claim 9, wherein the light compensating component is Micro light-emitting diodes (LEDs), and the light-emitting component is LEDs, Mini LEDs or Micro LEDs.

11. The tiled apparatus of claim 1, wherein the backlight module is a straight down backlight module, which comprises a light-emitting component and a diffusion plate, and the diffusion plate is disposed on a side of the light-emitting component close to the display panel; and wherein in a top view direction of the tiled apparatus, the light-emitting component overlaps with the display part, and the diffusion plate extends to an area where the tiled part is located.

12. The tiled apparatus of claim 1, wherein the backlight module is a side entry backlight module comprising a light-emitting component and a light guide plate; and wherein in a top view direction of the tiled apparatus, the light guide plate extends from an area where the display part is located to an area where the tiled part is located.

13. A tiled apparatus, comprising at least two display devices tiled with each other, wherein a seam is defined between adjacent two display devices, and each display device comprises: a display panel comprising a display part and a tiled part located on a side of the display part, wherein tiled parts of the adjacent two display devices are adjacent to each other, and the tiled parts are composed of transparent materials; and a backlight module disposed on a side of the display panel away from a display surface, wherein an orthographic projection of the backlight module on the display panel covers the display part and at least a part of the tiled part, and light emitted from an area of the backlight module corresponding to the tiled part is capable of passing through the tiled part; wherein the tiled apparatus further comprises a lens part disposed on a side of the display panel away from the backlight module, the lens part covers adjacent two tiled parts and the seam, and the lens part partially overlaps with display parts of adjacent two display panels; wherein an

overlap size between the lens part and the display part is less than or equal to 1 mm.

14. A tiled apparatus, comprising at least two display devices tiled with each other, wherein a seam is defined between adjacent two display devices, and each display device comprises: a display panel comprising a display part and a tiled part located on a side of the display part, wherein tiled parts of the adjacent two display devices are adjacent to each other, and the tiled parts are composed of transparent materials; and a backlight module disposed on a side of the display panel away from a display surface, wherein an orthographic projection of the backlight module on the display panel covers the display part and at least a part of the tiled part, and light emitted from an area of the backlight module corresponding to the tiled part is capable of passing through the tiled part; wherein the tiled apparatus further comprises a lens part disposed on a side of the display panel away from the backlight module, the lens part covers adjacent two tiled parts and the seam, and the lens part partially overlaps with display parts of adjacent two display panels; wherein the tiled part comprises a frame adhesive disposed at a boundary of the display panel and a support part disposed between the frame adhesive and the display part; and wherein the support part comprises a support body and a first main spacer column disposed on the support body, and in a top view direction of the tiled apparatus, an area of the first main spacer column is smaller than an area of the support body.

15. The tiled apparatus of claim 14, wherein the display part further comprises a second main spacer column, and an absolute value of difference between a thickness of the first main spacer column and a thickness of the second main spacer column ranges from 0 to 0.2 microns.

16. The tiled apparatus of claim 15, wherein the thickness of the first main spacer column is equal to the thickness of the second main spacer column.

17. The tiled apparatus of claim 14, wherein the seam between the adjacent two display devices is filled with a transparent optical adhesive.

18. The tiled apparatus of claim 17, wherein an absolute value of difference between a light transmittance of the frame adhesive and a light transmittance of the transparent optical adhesive ranges from 0 to 5%.

19. The tiled apparatus of claim 17, wherein a refractive index of the transparent optical adhesive is same as a refractive index of a substrate of the display panel.
